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GTIIT

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TUTORIAL 13 - JANUARY 7

POWER SERIES

GEOMETRIC SERIES

Is the sum of the powers of a given number $r \in \mathbb{R}$ and only convergent for $|r| < 1$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} : |r| < 1$$

$$\sum_{n=k}^{\infty} r^n = \frac{r^k}{1-r} : |r| < 1$$

The last is obtained from the first:

$$\sum_{n=k}^{\infty} r^n = r^k \cdot \sum_{n=0}^{\infty} r^n = r^k \frac{1}{1-r}$$

POWER SERIES

Definition. A power series is a series of the form

$$f(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + \dots + c_n(x - x_0)^n + \dots$$

where x is a variable and the c_n are the coefficients.

- For each fixed x , the series is a numerical series, that we can **test for convergence or divergence**.
- The result of the series is a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x - x_0)^n$$

whose **domain is the set of all x for which the series converges**.

- The advantage of representing a function f as a power series is that the study of f will be similar to a polynomial.

EXERCISE 1. GEOMETRIC SERIES.

Find expressions as power series on $(-1, 1)$:

a) $f(x) = \frac{1}{1-x^2}$

b) $g(x) = \frac{x}{(1-x^2)^2}$

Solution. Use a geometric series $\sum_{n=0}^{\infty} r^n$ convergent only for $|r| < 1$:

$$f(x) = \frac{1}{1-x^2} = \sum_{n=0}^{\infty} x^{2n} = 1 + x^2 + x^4 + \dots + x^{2n} : |x| < 1$$

$$\begin{aligned} g(x) &= \frac{x}{(1-x^2)^2} = \frac{1}{2} \frac{d}{dx} \left(\frac{1}{1-x^2} \right) = \frac{1}{2} \frac{d}{dx} \left(\sum_{n=0}^{\infty} x^{2n} \right) \\ &= \frac{1}{2} \sum_{n=1}^{\infty} 2n x^{2n-1} = \sum_{n=1}^{\infty} n x^{2n-1} : |x| < 1 \checkmark \end{aligned}$$

POWER SERIES

EXERCISE 2. Find all the values of x such that the series is convergent

$$f(x) = \sum_{n=0}^{\infty} \frac{(2x-5)^n}{\sqrt{n+2}}$$

- $a_n = \frac{(2x-5)^n}{\sqrt{n+2}}$ so Ratio test says $L < 1$ implies convergence:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(2x-5)^{n+1}}{\sqrt{n+3}}}{\frac{(2x-5)^n}{\sqrt{n+2}}} \right| = \lim_{n \rightarrow \infty} \frac{\sqrt{n+2}}{\sqrt{n+3}} |2x-5| = |2x-5| = L$$

⇒ it **converges for $|x - \frac{5}{2}| < \frac{1}{2}$** and **diverges for $|x - \frac{5}{2}| > \frac{1}{2}$:**

- The series **converges on $(2, 3)$**
- The series **diverges on $(-\infty, 2) \cup (3, \infty)$**
- The values for which $|2x - 5| = 1$ must be analyzed separately.

POWER SERIES

ENDPOINTS OF THE INTERVAL: $x = 3$

- For $2x - 5 = 1 \iff x = 3$

$\implies$ the series is an **divergent positive series**:

$$f(3) = \sum_{n=0}^{\infty} \frac{(2x-5)^n}{\sqrt{n+2}} \Big|_{x=3} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+2}} = \infty \checkmark$$

POWER SERIES

ENDPOINTS OF THE INTERVAL: $x = 2$

• For $2x - 5 = -1 \iff x = 2 \implies$ the series is an alternating convergent series with

• $b_n = \frac{1}{\sqrt{n+2}} > 0$ decreasing,

• $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+2}} = 0$:

$$\implies f(2) = \sum_{n=0}^{\infty} \frac{(2x-5)^n}{\sqrt{n+2}} \Big|_{x=2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+2}} \text{ convergent}$$

Hence, the power series converges on $[2, 3)$ ✓

POWER SERIES: INTERVAL OF CONVERGENCE

THEOREM A. For a given power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ there are only three possibilities:

- ① The series converges only when $x = a$
 - ② The series converges for all $x \in \mathbb{R}$
 - ③ There is $R > 0$ such that the series converges if $|x - a| < R$ and diverges if $|x - a| > R$. ✓
- The number $R > 0$ is called the **radius of convergence**.
 - In cases 1 and 2, let $R = 0$ and $R = \infty$, respectively.
 - The **interval of convergence** consists of all x for which the series converges. It can be:
 $(a - R, a + R)$, $[a - R, a + R)$, $(a - R, a + R]$ or $[a - R, a + R]$.

Ex 3. Find all the $x \in \mathbb{R}$ such that the series converges and the result of the infinite sum, when possible:

a) $\sum_{n=1}^{\infty} \left(\frac{1}{1+x^2} \right)^n$

b) $\sum_{n=3}^{\infty} e^{-nx}$

c) $\sum_{n=1}^{\infty} \frac{n^2}{2^n} (x-1)^n$

d) $\sum_{n=0}^{\infty} \left(\frac{1}{3-x^2} \right)^n$

Solutions. In a, b and d, use the condition of convergence of a geometric series $\sum_{n=k}^{\infty} r^n = \frac{r^k}{1-r}$ which is $|r| < 1$

a) For $r = \frac{1}{1+x^2}$, the sum and the intervals of convergence are:

$$\sum_{n=1}^{\infty} \left(\frac{1}{1+x^2} \right)^n = \frac{r^1}{1-r} = \frac{1}{x^2} \text{ on } I = (-\infty, 0) \cup (0, +\infty) \quad \checkmark$$

For a, b and d, use the condition of convergence of a geometric series, which is $|r| < 1$

Solutions. The sum and the intervals of convergence are:

b) For $r = e^{-x} \implies \sum_{n=3}^{\infty} e^{-nx} = \frac{e^{3x}}{1-e^x}$ on $I = (0, \infty)$

c) This not a geometric series and we apply the Ratio Test to get $R = 2$ and so, $I = (-1, 3)$.

d) $\sum_{n=0}^{\infty} \left(\frac{1}{3-x^2} \right)^n = \frac{x^2-3}{x^2-2}$ for all x such that

$$\left| \frac{1}{3-x^2} \right| < 1 \iff 1 < |3-x^2| \iff |x| < \sqrt{2} \text{ or } |x| > 2, \text{ so the series is convergent only for}$$

$$x \in (-\infty, -2) \cup (-\sqrt{2}, \sqrt{2}) \cup (2, +\infty)$$

POWER SERIES

Ex 4. a) Analyze the convergence of the series

$$\sum_{n=1}^{\infty} \left(\frac{9n+5}{4n+6} \right)^n$$

b) Find all the $x \in \mathbb{R}$ such that the series converges:

$$\sum_{n=1}^{\infty} (-1)^n \left(\frac{9n+5}{4n+6} \right)^n (x-3)^{2n}$$

Solution. a) The series $\sum_{n=1}^{\infty} \left(\frac{9n+5}{4n+6} \right)^n$ is divergent because of the Root Test, with limit $L > 1$:

$$\begin{aligned}\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} &= \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n+5}{4n+6} \right)^n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{9n+5}{4n+6} \right) = \frac{9}{4} > 1\end{aligned}$$

So, the series $\sum_{n=1}^{\infty} \left(\frac{9n+5}{4n+6} \right)^n$ is **divergent**.

b) Apply again the Root Test and solve for x imposing the condition that the limit $L < 1$:

$$\lim_{n \rightarrow \infty} \sqrt[n]{|A_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n+5}{4n+6}\right)^n |x-3|^{2n}}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{9n+5}{4n+6}\right) |x-3|^2 = \frac{9}{4} |x-3|^2 < 1 \iff |x-3| < \frac{2}{3}$$

- The series $\sum_{n=1}^{\infty} (-1)^n \left(\frac{9n+5}{4n+6}\right)^n (x-3)^{2n}$

is absolutely convergent for $x \in (\frac{7}{3}, \frac{11}{3})$

- The series diverges for $x \in (-\infty, \frac{7}{3}) \cup (\frac{11}{3}, \infty)$.

- The values $|x - 3| = \frac{2}{3}$ deserves a separate analyzes, and the series **diverges**, as we prove as follows. In this case

$$|x - 3| = \frac{2}{3} \implies |x - 3|^{2n} = \left(\frac{4}{9}\right)^n$$

So the series has the form

$$\sum_{n=1}^{\infty} \left(\frac{9n+5}{(4n+6)}\right)^n |x - 3|^{2n} = \sum_{n=1}^{\infty} \left(\frac{9n+5}{(4n+6)}\right)^n \left(\frac{4}{9}\right)^n$$

Notice that the Ratio Test is inconclusive:

$$\implies \lim_{n \rightarrow \infty} \sqrt[n]{|A_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n+5}{4n+6}\right)^n |x - 3|^{2n}} = \lim_{n \rightarrow \infty} \left(\frac{9n+5}{4n+6}\right) \frac{4}{9} = 1$$

Prove that the general term does not tend to 0

$$\begin{aligned}\lim_{n \rightarrow \infty} |A_n| &= \lim_{n \rightarrow \infty} \left(\frac{9n+5}{(4n+6)} \right)^n \left(\frac{4}{9} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{36n+20}{36n+54} \right)^n \\&= \lim_{n \rightarrow \infty} \left(\frac{18n+10}{18n+27} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{18n+27-17}{18n+27} \right)^n \\&= \lim_{n \rightarrow \infty} \left(1 - \frac{17}{18n+27} \right)^n \\&= \lim_{n \rightarrow \infty} \left[\overbrace{\left\{ \left(1 - \frac{17}{18n+27} \right)^{-\frac{18n+27}{17}} \right\}}^{\rightarrow e} \right]^{-\frac{17n}{18n+27}} = e^{-\frac{17}{18}} \neq 0\end{aligned}$$

Then $\lim_{n \rightarrow \infty} |A_n| \neq 0$

We've proved that

$$\lim_{n \rightarrow \infty} |a_n| \neq 0 \implies \lim_{n \rightarrow \infty} a_n \neq 0$$

Then,

$$\sum_{n=1}^{\infty} |a_n| \text{ and } \sum_{n=1}^{\infty} (-1)^n a_n \text{ are both divergent series } \checkmark$$

$$\text{Hence, } \sum_{n=1}^{\infty} \left(\frac{9n+5}{(4n+6)} \right)^n \left(\frac{4}{9} \right)^n \text{ and } \sum_{n=1}^{\infty} (-1)^n \left(\frac{9n+5}{(4n+6)} \right)^n \left(\frac{4}{9} \right)^n$$

are divergent series. $\checkmark$

Ex. 5. Find the Taylor Series

a) $f(x) = e^x$ at $x_0 = 1$,

c) $g(x) = \sin^2(x)$ at $x = 0$

b) $h(x) = e^x$ at $x_0 = 4$

ution. • In a, b and c), the interval of convergence is

$$(-\infty, +\infty) = \mathbb{R}.$$

• Use the known series of the function center at $x_0 = 0$:

a) $\forall x \in \mathbb{R}, f(x) = e^x = e \cdot e^{x-1} = e \cdot \sum_{n=0}^{\infty} \frac{1}{n!} (x-1)^n$

b) $\forall x \in \mathbb{R}, h(x) = e^x = e^4 \cdot e^{x-4} = e^4 \cdot \sum_{n=0}^{\infty} \frac{1}{n!} (x-4)^n$

Find the Taylor Series

$$\begin{aligned} \text{c) } \forall x \in \mathbb{R}, g(x) = \sin^2(x) &= \frac{1 - \cos(2x)}{2} = \frac{1}{2} - \frac{1}{2} \cos(2x) \\ &= \frac{1}{2} - \frac{1}{2} \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (2x)^{2n} \right] = \frac{1}{2} - \frac{1}{2} - \frac{1}{2} \left[\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} (2x)^{2n} \right] \end{aligned}$$

$$= -\frac{1}{2} \left[\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} 2^{2n} x^{2n} \right]$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n)!} 2^{2n-1} x^{2n}$$

REPRESENTATION OF FUNCTIONS AS POWER SERIES

EXERCISE 6. Represent $f(x) = \frac{1}{1-x}$, $g(x) = \frac{1}{(1-x)^2}$, $h(x) = \frac{(1+x)^4}{(1-x)^2}$ as series on $(-1-R, -1+R)$

Solution: • f is the the result of a geometric series centered at $x = -1$:

$$\begin{aligned} f(x) &= \frac{1}{1-x} = \frac{1}{2-(x+1)} = \frac{1}{2} \frac{1}{\left[1 - \left(\frac{x+1}{2}\right)\right]} = \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{(x+1)}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (x+1)^n : |x+1| < 2 \end{aligned}$$

- The series converges on $(-3, 1)$
- The series diverges on $(-\infty, -3) \cup (1, \infty)$
- The values for which $|x+1| = 2$: the series diverges because it is a geometric series.

Check the divergence for the values such that $|x + 1| = 2$:

- For $x = -3 \Rightarrow \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (x + 1)^n \Big|_{x=-3}$
$$= \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (-2)^n = \sum_{n=0}^{\infty} \frac{1}{2} (-1)^n$$

which is divergent

- For $x = 1 \Rightarrow \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (x + 1)^n \Big|_{x=1}$
$$= \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (2)^n = \sum_{n=0}^{\infty} \frac{1}{2}$$

which is divergent as well ✓

- $$\begin{aligned}
 g(x) &= \frac{1}{(1-x)^2} \\
 &= \frac{d}{dx} \left[\frac{1}{1-x} \right] = \frac{d}{dx} \left[\sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (x+1)^n \right] = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \left[\frac{d}{dx} (x+1)^n \right] \\
 &= \sum_{n=1}^{\infty} \frac{1}{2^{n+1}} [n \cdot (x+1)^{n-1}] = \sum_{k=0}^{\infty} \frac{1}{2^{k+2}} (k+1) \cdot (x+1)^k
 \end{aligned}$$

- Geometric series converges for $|x+1| < 2$ and it diverges for $|x+1| \geq 2$

$\implies R = 2$ and the interval of convergence is $(-3, 1)$ ✓

- $$h(x) = \frac{(1+x)^4}{(1-x)^2} = (1+x)^4 \frac{1}{(1-x)^2} = (1+x)^4 \cdot h(x)$$

Hence, with the same interval of convergence, the series of h is obtained by multiplying the series of g founded in the previous page:

$$\begin{aligned} h(x) &= (1+x)^4 \frac{1}{(1-x)^2} = (1+x)^4 \cdot h(x) = \sum_{k=0}^{\infty} \frac{1}{2^{k+2}} (k+1) \cdot (x+1)^{k+4} \\ &= \sum_{j=4}^{\infty} \frac{1}{2^{j-2}} (j-3) \cdot (x+1)^j : |x+1| < 2 \checkmark \end{aligned}$$

- Geometric series implies the convergence for $|x+1| < 2$ and the divergence for $|x+1| \geq 2$

$\Rightarrow R = 2$ and the interval of convergence is $(-3, 1) \checkmark$

REPRESENTATION OF FUNCTIONS AS POWER SERIES

Ex 7. Represent $h(x) = \frac{x^4}{(1-x)}$ as power series on $(-1, 1)$

Solution: Find a relationship with a known function

$$\begin{aligned} \bullet h(x) &= \frac{x^4}{1-x} = x^4 \cdot \left(\frac{1}{1-x} \right) \\ &= x^4 \cdot \left(\sum_{n=0}^{\infty} x^n \right) = \sum_{n=0}^{\infty} x^{n+4} \end{aligned}$$

Let $k = n + 4 \implies 4 \leq k$

$$\implies h(x) = \sum_{k=4}^{\infty} x^k : |x| < 1 \checkmark$$

DIFFERENTIATION AND INTEGRATION OF POWER SERIES.

B. TAYLOR THEOREM. If the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, then the function

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

is differentiable (and continuous) on $(a-R, a+R)$ with

$$\textcircled{1} \quad f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots$$

$$= \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$$

$$\textcircled{2} \quad \int f(x)dx = C + c_0(x-a) + c_1 \frac{(x-a)^2}{2} + c_2 \frac{(x-a)^3}{3} + \dots$$
$$= C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{(n+1)}$$

$\textcircled{3}$ Both series converge $\forall |x-a| < R$ and diverge $\forall |x-a| > R$

TAYLOR SERIES

The previous and the following theorem are also true if $R = \infty$, for which the interval of convergence is

$$(-\infty, \infty) \checkmark$$

TAYLOR SERIES

DEFINITION AND THEOREM C. If f has a power series representation centered at a , then for $|x - a| < R$:

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

and the coefficients are given by

$$c_n = \frac{f^{(n)}(a)}{n!}$$

where $f^{(n)}(a)$ denotes the value of the n -th derivative of f evaluated at $x = a$. So, for $|x - a| < R$,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

DEFINITION AND THEOREM D. If f has a power series representation centered at $a = 0$, then for $|x| < R$:

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots$$

and its coefficients are given by

$$c_n = \frac{f^{(n)}(0)}{n!}$$

where $f^{(n)}(0)$ denotes the value of the n -th derivative of f evaluated at $a = 0$. The Taylor Series is called Maclaurin series and for all $|x| < R$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$$

MACLAURIN SERIES

Exercise 8. Find the Maclaurin series for

a) $f(x) = e^{-x^2}$

b) $F(x) = \int e^{-x^2} dx$

c) Compute $\int_0^1 e^{-x^2} dx$ with accuracy of 10^{-3} .

Maclaurin Series 8. a)

- The Theorem implies that $g(t) = e^t$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!} \text{ for all } x \in \mathbb{R}$$

- Use the Taylor series for $g(t) = e^t$ and compose with $t = -x^2 \forall t \in \mathbb{R}$:

$$e^{-x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!} \text{ for all } x \in \mathbb{R} \checkmark$$

Antiderivative of the previous function

b) Taylor Theorem allows us to integrate on the interval of convergence, which is all $\mathbb{R}$, to find the Taylor series of the antiderivative:

$$\int e^{-x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \int x^{2n} dx = C + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)} x^{2n+1} \quad \forall x \in \mathbb{R} \quad \checkmark$$

Ex 8. c) Apply the Fundamental Theorem of Calculus

$$\int_0^1 e^{-x^2} dx = \left[C + \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)} x^{2n+1} \right] \Big|_{x=0}^{x=1} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} \frac{1}{(2n+1)}$$

$$= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots$$

We know by the Alternating Series Estimation Theorem

$$0 \leq |R_n| \leq |a_{n+1}| \leq \frac{1}{(n+1)!(2n+1)} < 10^{-3} \text{ for } n \geq 5$$

because $n = 5 \implies 5!(2n+1) = 120 \cdot 11 > 1200 > 10^3$ ✓

Hence, let us approximate the definite integral with the 5 first terms of the series:

$$\int_0^1 e^{-x^2} dx \approx 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} \approx 0.7475 \checkmark$$

True Value: $\int_0^1 e^{-x^2} dx \approx 0.7468 \implies |R_5| = |0.7475 - 0.7468| \approx 0.0007 < 0.001 \checkmark$

Exercise 9. Find the Taylor series for

a) $f(x) = \sin(x)$ at $a = \pi$.

b) $g(x) = \sin^2(x) = \frac{1}{2}(1 - \cos(2x))$ at $a = 0$

ution. a) Use the known series, that is convergent $\forall t \in \mathbb{R}$:

- $\sin(t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} t^{2n+1}$

- Use the identity and compose with $t = x - \pi$:

$$f(x) = \sin(x) = \sin(x - \pi) \cos(\pi)^{-1} + \cos(x - \pi) \sin(\pi)^0$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!} (x - \pi)^{2n+1} \quad \forall x \in \mathbb{R} \checkmark$$

MACLAURIN SERIES

$$\text{b) } g(x) = \sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

$$= \frac{1}{2} \cdot \left[1 - 1 - \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} (2x)^{2n} \right] \forall x \in \mathbb{R} \checkmark$$

$$= x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6 + \dots$$

$$= \sum_{n=1}^{\infty} 2^{2n-1} \frac{(-1)^{n+1}}{(2n)!} x^{2n} \forall x \in \mathbb{R} \checkmark$$

TAYLOR POLYNOMIAL AND TAYLOR SERIES

$$f(x) = \underbrace{f(a) + f'(a)(x-a) + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n}_{T_n(x)} + \underbrace{\sum_{k=n+1}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k}_{R_n(x)}$$

= Taylor Polynomial + the remainder $R_n(x)$

TAYLOR AND MACLAURIN SERIES

THEOREM E. Let $f(x)$ be a function that has derivatives of any order and $f(x) = T_n(x) + R_n(x)$, where T_n is the n th-degree

Taylor polynomial of f at a , $T_n[f(x)] = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$.

If $\lim_{n \rightarrow \infty} R_n(x) = 0$ for all $|x - a| < R$

then $f(x)$ is equal to the sum of its Taylor series on $(a - R, a + R)$, that is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \dots \checkmark$$

TAYLOR AND MACLAURIN SERIES

THEOREM F. TAYLOR INEQUALITY. Let $f(x)$ be a function that has derivatives of any order and $f(x) = T_n(x) + R_n(x)$, where T_n is the n th-degree Taylor polynomial of f at a ,

$$T_n[f(x)] = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

If there exist $M > 0$ and d such that

$$|f^{(n+1)}(x)| \leq M \text{ for all } |x - a| < d$$

then the remainder of the Taylor series satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x - a|^{n+1} \text{ for all } |x - a| < d$$

MACLAURIN SERIES

Lemma G. Prove that

$$\lim_{n \rightarrow \infty} \frac{1}{(n+1)!} |x|^n = 0 \text{ for all } x$$

Proof of the Lemma G.

Recall the following part of the Ratio test:

Let (a_n) be a sequence of non-zero real numbers. If the following limit exists and

$$\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = L < 1$$

then $\sum_n a_n$ is an absolutely convergent series. In particular, $\lim_{n \rightarrow \infty} a_n = 0$.

As a consequence, we can prove the following:

For any fixed real number x , the following sequence goes to zero:

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

Proof. We consider the sequence $a_n = \frac{x^n}{n!}$.

If $x = 0$ we get $a_n = 0$ for all $n \geq 1$, that clearly has limit equal to zero.

If $x \neq 0$ we use the Ratio test for the series $\sum a_n = \sum \frac{x^n}{n!}$:

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} &= \lim_{n \rightarrow \infty} \frac{|x^{n+1}|}{(n+1)!} \frac{n!}{|x^n|} \\ &= \lim_{n \rightarrow \infty} \frac{|x| \cancel{n!}}{(n+1) \cancel{n!}} = \lim_{n \rightarrow \infty} \frac{|x|}{(n+1)} = 0\end{aligned}$$

Since $L = 0 < 1$, we conclude that the series converges, and so, the general term tends to zero:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad \checkmark$$

Ex 10. a) Prove that for all $x \in \mathbb{R}$, $f(x) = e^x$ is equal to the sum of its Maclaurin series.

Solution. • All the derivatives are the same:

$$f(x) = e^x \implies f^{(n)}(x) = e^x \quad \forall n, x.$$

• If $d > 0$ is any positive number, then

$$|x| < d \implies |f^{(n+1)}(x)| = e^x < e^d$$

then, take $M = e^d$ and so obtain for the remainder

$$|R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1} \text{ for all } |x| < d$$

• Notice that the same constant $M = e^d$ works for every value of n .

10. a) We had obtained: $0 \leq R_n \leq \frac{e^d}{(n+1)!} |x|^{n+1}$ for all $|x| < d$

• Moreover, by the **Lemma G**:

$$\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x|^{n+1} = e^d \cdot \lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} = 0 \text{ for all } |x| < d$$

• By the Squeeze Theorem:

$$\lim_{n \rightarrow \infty} |R_n(x)| = 0 \implies \lim_{n \rightarrow \infty} R_n(x) = 0$$

• The Theorem implies that then $f(x) = e^x$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for all } x \in \mathbb{R} \checkmark$$

MACLAURIN SERIES

10. b) Example. Find the Taylor series for $f(x) = e^x$ centered at $x = \pi$ and the interval of convergence.

Solution. • All the derivatives are the same:

$$f(x) = e^x \implies f^{(n)}(2) = e^\pi \quad \forall n.$$

• We may continue in the same way and prove the convergence of the Taylor series at $a = 2$ as before, or:

$$e^x = e^{(x-\pi)+\pi} = e^{(x-\pi)} e^\pi$$

• Use the part a) of this example for $(x - \pi)$

So, as before, the Theorem implies that then $f(x) = e^x$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$e^x = \sum_{n=0}^{\infty} \frac{e^\pi}{n!} (x - \pi)^n \quad \text{for all } x \in \mathbb{R} \checkmark$$

MACLAURIN SERIES: REVIEW

Ex 11. Find the Maclaurin series $f(x) = \sin(x)$ and prove that it converges to f on $\mathbb{R}$.

Solution. • Arrange the computations in two columns as follows;
the derivatives are periodic in a cycle of 4:

- $f(x) = \sin(x) \implies f(0) = 0$

- $f'(x) = \cos(x) \implies f'(0) = 1$

- $f^{(2)}(x) = -\sin(x) \implies f^{(2)}(0) = 0$

- $f^{(3)}(x) = -\cos(x) \implies f^{(3)}(0) = -1$

- $f^{(4)}(x) = +\sin(x) \implies f^{(4)}(0) = 0$

$\implies$ the even derivatives are all zero, and the odd derivatives are ± 1

MACLAURIN SERIES OF $\sin(x)$

- $f^{(2n)}(0) = 0 \quad \forall n$
- $f^{(2n+1)}(0) = (-1)^n \quad \forall n$
- The Maclaurin series is then

$$f(x) = \sin(x) = f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

MACLAURIN SERIES OF $\sin(x)$.

$$|\sin(x)| \leq 1 \text{ and } |\cos(x)| \leq 1 \implies |f^{n+1}(x)| \leq 1 \text{ for all } x \in \mathbb{R}$$

$$\implies 0 \leq |R_n| \leq \frac{|f^{n+1}(x)|}{(n+1)!} |x|^{n+1} \leq \frac{1}{(n+1)!} |x|^{n+1} \text{ for all } x$$

- Moreover, by the **Lemma G**:

$$\lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} = 0 \text{ for all } x$$

- By the Squeeze Theorem:

$$\lim_{n \rightarrow \infty} |R_n(x)| = 0 \implies \lim_{n \rightarrow \infty} R_n(x) = 0$$

MACLAURIN SERIES OF $\sin(x)$.

- The Theorem implies that then $f(x) = \sin(x)$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad \text{for all } x \in \mathbb{R} \checkmark$$

MACLAURIN SERIES OF $\cos(x)$.

Ex 12. a) Find the Maclaurin series $g(x) = \cos(x)$ and prove that it converges to g on $\mathbb{R}$.

Solution. g is the derivative of $\sin(x)$:

$$\cos(x) = \frac{d}{dx} \sin(x)$$

, by the Theorem, derivate the previous series to obtain:

$$\cos(x) = \frac{d}{dx} \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \right] = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2n+1)x^{2n}$$

Attention that the series of the derivative **starts at $n = 0$** , since **the derivative of its first term is $\frac{d}{dx}x = 1 \neq 0$** :

MACLAURIN SERIES OF $\cos(x)$.

- The Theorem implies that then $g(x) = \cos(x)$ is equal to the sum of its Maclaurin series on $\mathbb{R}$, so $R = \infty$:

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \text{ for all } x \in \mathbb{R} \checkmark$$

MACLAURIN SERIES OF $\cos(x)$.

Ex 12. b) Find the Maclaurin series $h(x) = x^3 \cdot \cos(x)$ and prove that it converges to h on $\mathbb{R}$.

Solution. The theorem allows to multiply x^3 by the series, which will be convergent to the function for all $x \in (a - R, a + R)$, hence, in this case, for all $x \in \mathbb{R}$:

$$\begin{aligned} h(x) &= x^3 \cdot \cos(x) = x^3 \cdot \left[\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \right] \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+3} = x^3 - \frac{1}{2}x^5 + \frac{1}{24}x^7 - \dots \end{aligned}$$

Let $k = n + 1$, so the series is equal to

$$h(x) = x^3 \cdot \cos(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2(k-1))!} x^{2k+1} \quad \checkmark$$